

1. THE PROBLEM (2-HOUR DISPATCH CLIFF & CASCADING SUPPLY BIAS)

- 🚨Morning Dropout Crisis: Home cooks drop out unexpectedly at 10:30 AM while lunch dispatch locks at 12:30 PM (a hard 120-min operational cutoff across Bengaluru, Mumbai, and Delhi).
- 🚨Silent Dropout Blindspot: Unstructured Hinglish WhatsApp messages (e.g., Sunita Kulkarni CK090 at 7:41 AM) are missed in standups, leaving the sheet active while 6 customers wait.
- 🚨Cascading Supply Depletion: Sequential greedy matching exhausts scarce Jain-certified kitchens on flexible vegetarians, stranding subsequent strict Jain subscribers.

2. TARGET PERSONAS & CORE WORKFLOWS

- 👤Primary (City Ops Manager): Under intense morning time pressure; needs to resolve multi-cook dropouts across 500+ subscribers within 10 minutes without making calculation errors.
- 👤Secondary (Head of Ops / VP): Oversees city unit economics; needs visibility into network fragility heatmaps, cook reliability rankings, and net refund leakage to manage supply chain.

3. SUCCESS METRICS & GUARDRAILS (GROUNDED IN 7,138 HISTORICAL ORDERS)

Metric Category	Production Metric & Exact Formula	Historical Baseline	Production SLA
North Star Outcome	Meal Recovery Rate = (Reassigned Meals Delivered / Total Impacted Meals) * 100	71.4% (Unmanaged)	>= 88.0%
Operational Velocity	Time-to-Resolution (TTR): Elapsed time from dropout detection to batch dispatch	42 mins (Manual phone/sheet)	<= 8 mins (by 10:45 AM)
Dietary Integrity	Zero-Tolerance Diet Mismatches: Unintended cross-diet assignments (Jain to non-Jain)	3.2% Incident exposure	0.0% (Hard Gate)
Financial Guardrail	Monthly Net Refund Leakage: Cash refunded due to avoidable cook cancellations	Rs 21,480 / month	<= Rs 12,000 / month
Subscriber Retention	14-Day Post-Incident Churn: % of subscribers cancelling within 14 days of incident	28.5% (Cancelled meals)	<= 6.0% for rescued meals

4. WHAT OUR PROTOTYPE PROVED (EMPIRICAL VALIDATION)

- 🧠Hinglish Parsing Ingests Ground Truth: Natural language parser accurately caught Sunita Kulkarni's silent dropout (CK090), preventing 6 lunch delivery failures in Mumbai.
- 🧠Global Scarcity-First Knapsack Works: Solving all morning dropouts simultaneously prioritized Jain meals first, saving 20% more meals than sequential greedy queue matching.
- 🧠Explainable AI Drives Velocity: Transparent match scores (City v | Diet v | Spare capacity v | Cuisine v) enabled 1-click operator approval for 24 affected orders under 10 seconds.
- 🧠Dynamic Capacity Boundaries Enforced: Strictly respected max_daily_orders, gracefully triggering proactive refunds + Rs 50 wallet credits when kitchens reached ceiling.

5. CORE ARCHITECTURAL DECISIONS: WHAT WE DELIBERATELY CUT & WHY

- 🚫Two-Way Cook Negotiation Chat: Cut: Asking backup cooks to accept/decline emergency meals via WhatsApp. Why: Cook response latency exceeds 25 mins at 10:30 AM, guaranteeing late meals.
- 🚫Subscriber Meal Swapping / Approval: Cut: Allowing subscribers to pick alternate dishes or substitute cooks. Why: Working professionals take >45 mins to reply during office hours. Replacement logic is too complex.
- 🚫Dynamic Surge Bidding for Cooks: Cut: Real-time surge pricing auctions for backup cooks. Why: Creates adverse incentives where cooks intentionally delay availability to capture surge bonuses.
- 🚫Autonomous "Zero-Click" Auto-Dispatch: Cut: Having AI cancel/reassign orders directly without human review. Why: WhatsApp messages contain sarcasm, family members chatting, and high emotional stakes.

6. OPEN QUESTIONS BEFORE ENGINEERING BUILD

- 🔍Multi-Operator Concurrency: When 2 ops managers resolve dropouts in the same city simultaneously, what distributed locking mechanism (Redis vs Postgres row locks) prevents double-bookings?
- 🔍Kitchen Certification Protocol: What operational inspection standard elevates a cook from "self-declared Jain" to "verified separate Jain kitchen" to eliminate religious liability?
- 🔍Cook Acceptance SLA Timeout: What automated fallback sequence executes if an assigned backup cook fails to acknowledge the emergency batch dispatch within 12 minutes of assignment?
- 🔍WhatsApp Business API Sandbox: Are pre-approved Meta WhatsApp utility message templates active for dynamic refund amounts, backup cook names, and real-time delivery ETAs?